🏟️ Turf Booking System - SQL Case Study

📍 Introduction

This comprehensive SQL case study simulating a **real-world turf booking system**. This database design demonstrates how SQL can be used to build a production-ready booking platform with features like dynamic pricing, wallet management, referral systems, and automated business logic.

The system handles complete operations including user management, turf bookings, payments, wallet transactions, referral rewards, offers/discounts, and analytics reporting.

🎯 Business Objectives

- Track revenue and booking performance across different turfs
- Implement dynamic pricing (weekday vs weekend)
- Manage user wallets and transaction history
- Run referral programs with automated rewards
- Apply time-based offers and discounts
- Prevent double bookings with triggers
- Analyze customer behavior and retention
- Generate business insights through SQL queries

🗄️ Database Schema Overview

The database consists of **14 core tables** representing real-world booking operations:

Table	Purpose
`roles`	Stores different user roles in the system (admin, user, owner)

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`users`	Stores login and basic profile information for all users
`turfs`	Stores details of sports grounds available for booking
`amenities`	Stores facilities offered at turfs (Floodlights, Parking, etc.)
`turf_amenities`	Maps amenities to specific turfs (many-to-many relationship)
`slots`	Stores time slots for each turf (1-hour or 2-hour durations)
`pricing_rules`	Defines weekday and weekend pricing for each turf
`offers`	Stores discount campaigns with validity date ranges
`bookings`	Tracks customer reservations, pricing, and status
`payments`	Stores payment transactions made by users
`wallets`	Stores current wallet balance for each user
`wallet_transactions`	Tracks all credit/debit activities in user wallets
`referrals`	Tracks referral relationships between users and reward status
`reviews`	Stores customer ratings and feedback for turfs

📊 Table Structures

1. `roles` – User Role Definitions

Purpose: Stores different user roles in the system.

Column	Type	Description
`role_id`	INT PRIMARY KEY	Unique role identifier
`role_name`	ENUM	Values: 'admin', 'user', 'owner'

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2. `users` – User Accounts

Purpose: Stores login and basic profile information for all users.

Column	Type	Description
`user_id`	INT PRIMARY KEY	Unique user identifier
`name`	VARCHAR(100)	Full name of the user
`phone`	VARCHAR(15) UNIQUE	Contact number

Column	Type	Description
`password`	VARCHAR(255)	Encrypted password
`email`	VARCHAR(100) UNIQUE	Email address
`role_id`	INT FOREIGN KEY	References `roles` table
`referred_by`	INT FOREIGN KEY	References `users` (who referred this user)
`referral_code`	VARCHAR(20) UNIQUE	Unique code for sharing
`created_at`	TIMESTAMP	Account creation timestamp

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3. `turfs` – Sports Grounds

Purpose: Stores details of sports grounds available for booking.

Column	Type	Description
`turf_id`	INT PRIMARY KEY	Unique turf identifier
`owner_id`	INT FOREIGN KEY	References `users` (turf owner)
`name`	VARCHAR(100)	Turf display name
`location`	VARCHAR(255)	Address/Landmark
`description`	TEXT	Additional details
`created_at`	TIMESTAMP	Record creation time

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4. `amenities` – Facilities

Purpose: Stores facilities offered at turfs (Floodlights, Parking, etc.).

Column	Type	Description
`amenity_id`	INT PRIMARY KEY	Unique amenity identifier
`name`	VARCHAR(100)	Amenity name (Floodlights, Parking, etc.)

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5. `turf_amenities` – Turf-Amenity Mapping

Purpose: Maps amenities to specific turfs (many-to-many relationship).

Column	Type	Description
`turf_id`	INT FOREIGN KEY	References `turfs`

Column	Type	Description
`amenity_id`	INT FOREIGN KEY	References `amenities`

Primary Key: `(turf_id, amenity_id)`

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6. `slots` – Time Slots

Purpose: Stores time slots for each turf (1-hour or 2-hour durations).

Column	Type	Description
`slot_id`	INT PRIMARY KEY	Unique slot identifier
`turf_id`	INT FOREIGN KEY	References `turfs`
`start_time`	TIME	Starting time (e.g., 06:00:00)
`end_time`	TIME	Ending time (e.g., 07:00:00)

Example: Football turf has 1-hour slots (6 AM to 10 PM), Cricket turf has 2-hour slots.

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7. `pricing_rules` – Dynamic Pricing

Purpose: Defines weekday and weekend pricing for each turf.

Column	Type	Description
`rule_id`	INT PRIMARY KEY	Unique rule identifier
`turf_id`	INT FOREIGN KEY	References `turfs`
`day_type`	ENUM	Values: 'weekday', 'weekend'
`price`	DECIMAL(10,2)	Price per slot

Business Logic: Weekend prices are higher due to increased demand.

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8. `offers` – Discount Campaigns

Purpose: Stores discount campaigns with validity date ranges.

Column	Type	Description
`offer_id`	INT PRIMARY KEY	Unique offer identifier
`offer_name`	VARCHAR(100)	Display name (e.g., "Monsoon Special")

Column	Type	Description
`discount_percent`	INT	Discount percentage (e.g., 20)
`valid_from`	DATE	Offer start date
`valid_to`	DATE	Offer end date
`turf_id`	INT NULL	Specific turf or NULL for all turfs

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9. `bookings` – Customer Reservations

Purpose: Tracks customer reservations, pricing, and status.

Column	Type	Description
`booking_id`	INT PRIMARY KEY	Unique booking identifier
`user_id`	INT FOREIGN KEY	References `users`
`slot_id`	INT FOREIGN KEY	References `slots`
`booking_date`	DATE	Date of the booking
`final_price`	DECIMAL(10,2)	Price after applying offers
`status`	ENUM	'pending', 'confirmed', 'cancelled'
`offer_id`	INT FOREIGN KEY	References `offers` (which offer was applied)
`created_at`	TIMESTAMP	Booking creation time

Unique Constraint: `(slot_id, booking_date)` – Prevents double booking.

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10. `payments` – Transaction Records

Purpose: Stores payment transactions made by users.

Column	Type	Description
`payment_id`	INT PRIMARY KEY	Unique payment identifier
`booking_id`	INT FOREIGN KEY	References `bookings`
`amount`	DECIMAL(10,2)	Amount paid
`status`	VARCHAR(50)	'paid', 'failed', 'pending'
`payment_method`	ENUM	'UPI', 'WALLET', 'CARD'
`created_at`	TIMESTAMP	Payment timestamp

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11. `wallets` – User Wallet Balances

Purpose: Stores current wallet balance for each user.

Column	Type	Description
`wallet_id`	INT PRIMARY KEY	Unique wallet identifier
`user_id`	INT UNIQUE FOREIGN KEY	References `users` (one-to-one)
`amount`	INT DEFAULT 0	Current wallet balance

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12. `wallet_transactions` – Wallet Ledger

Purpose: Tracks all credit/debit activities in user wallets.

Column	Type	Description
`txn_id`	INT PRIMARY KEY	Unique transaction identifier
`user_id`	INT FOREIGN KEY	References `users`
`amount`	DECIMAL(10,2)	Transaction amount
`type`	ENUM	'credit' or 'debit'
`description`	VARCHAR(255)	Transaction description
`created_at`	TIMESTAMP	Transaction timestamp

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13. `referrals` – Referral Tracking

Purpose: Tracks referral relationships between users and reward status.

Column	Type	Description
`referral_id`	INT PRIMARY KEY	Unique referral identifier
`referrer_id`	INT FOREIGN KEY	Person who referred
`referred_user_id`	INT FOREIGN KEY	Person who signed up
`reward_amount`	INT DEFAULT 0	Reward amount (₹100)
`status`	ENUM	'pending' or 'rewarded'

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14. `reviews` – Customer Feedback

Purpose: Stores customer ratings and feedback for turfs.

Column	Type	Description
`review_id`	INT PRIMARY KEY	Unique review identifier
`user_id`	INT FOREIGN KEY	References `users`
`turf_id`	INT FOREIGN KEY	References `turfs`
`rating`	INT	1 to 5 rating
`comment`	TEXT	Review text
`created_at`	TIMESTAMP	Review timestamp

⚙️ Automated Triggers

The database uses **6 triggers** to automate business logic:

Trigger Name	Event	Purpose
`after_signup_wallet`	AFTER INSERT ON users	Creates wallet for new user
`after_signup_bonus`	AFTER INSERT ON users	Gives ₹50 signup bonus for referred users
`before_booking_check`	BEFORE INSERT ON bookings	Prevents double booking
`before_booking_price`	BEFORE INSERT ON bookings	Calculates final price with weekday/weekend pricing + offers
`before_payment_balance`	BEFORE INSERT ON payments	Checks wallet balance for wallet payments
`after_payment_process`	AFTER INSERT ON payments	Confirms booking, deducts wallet, gives referral reward

⏰ Scheduled Events

Event Name	Schedule	Purpose
`event_cancel_unpaid`	EVERY 1 MINUTE	Auto-cancels pending bookings older than 10 minutes

📈 Business Intelligence Queries

1. What is the total revenue generated by the institute?

Insight: Overall business health and financial performance KPI

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2. Which turfs generate the highest revenue?

Insight: Identifies star performers vs underperforming turfs for investment decisions

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3. How much more revenue do weekends generate compared to weekdays?

Insight: Quantifies weekend premium to justify dynamic pricing strategy

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4. Who are the top 3 customers who spent the most money?

Insight: Identifies VIP customers for loyalty programs and special offers

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5. Which hours of the day get the most bookings?

Insight: Reveals peak demand hours for staffing and pricing optimization

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6. Are offers bringing more bookings? Which offer is most popular?

Insight: Measures campaign effectiveness and helps decide which discounts to continue

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7. How many customers came from referrals? How much revenue did they generate?

Insight: Calculates referral program ROI and shows if investment is paying off

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8. How many customers come back for a second, third, or fourth booking?

Insight: Measures customer loyalty and retention rate

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9. What percentage of bookings get cancelled vs confirmed vs pending?

Insight: Shows booking funnel health and identifies cancellation problems

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10. How do customers prefer to pay? (UPI, Wallet, or Card)

Insight: Reveals payment method preferences for gateway strategy and wallet promotion

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11. Which hours have low bookings and could benefit from promotions?

Insight: Identifies off-peak hours where discounts can drive additional revenue

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12. How many customers use wallet? What's the total money stored?

Insight: Shows wallet adoption rate and total locked-in customer value

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13. Which day of the week has the most confirmed bookings?

Insight: Helps plan weekly staffing, maintenance, and targeted promotions

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14. Who are our most valuable customers and what are their spending tiers?

Insight: Enables customer segmentation (Platinum/Gold/Silver/Bronze) for targeted marketing

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15. Do offers actually increase bookings or just reduce average price?

Insight: Compares with/without offer to determine if discounts drive volume or cannibalize revenue

🚀 Key Features Demonstrated

Feature	Implementation
Dynamic Pricing	`pricing_rules` table with weekday/weekend logic
Auto Discounts	`offers` table + `before_booking_price` trigger
Double Booking Prevention	Unique constraint + trigger
Wallet System	`wallets` + `wallet_transactions` + triggers
Referral Program	`referrals` + bonus/reward triggers
Payment Methods	UPI, Wallet, Card
Auto-Cancel Unpaid	MySQL Event (runs every minute)
Business Analytics	15 comprehensive SQL queries

✅ Outcome

This case study demonstrates how SQL can be used to build a **production-ready turf booking system** with:

- ✅ Automated business logic using triggers

- ✅ Scheduled events for maintenance
- ✅ Dynamic pricing and offers
- ✅ Wallet and referral systems
- ✅ Comprehensive analytics capabilities
- ✅ Real-world industry use cases